

Refilling the Buffer in a Storm

The Netherlands' countercyclical capital buffer, 0% → 2% (2022–2024)

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Micro- & Macroprudential Management · Country implementation of a CCyB increase

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STORY · what to say

- Good morning. Our case is the Netherlands.
- In 2022 and 2023 the Dutch central bank, De Nederlandsche Bank or DNB, raised its countercyclical capital buffer from zero to two percent. We want to explain why it did that, and whether it was a good idea.

LIKELY CHALLENGES · definitions and logic

Q What is the CCyB, in one sentence?

- A capital buffer of 0–2.5% of risk-weighted assets on domestic exposures that is built up in good times and released in bad times, so banks can absorb losses and keep lending.

Q Who decides it in the Netherlands?

- De Nederlandsche Bank (DNB), the Dutch central bank and designated macroprudential authority. It reviews the rate every quarter.

Q What exactly changed?

- 0% → 1% (announced May 2022, binding 25 May 2023) → 2% (announced 31 May 2023, binding 31 May 2024). It has been held at 2% since.

In 2022–23 the Netherlands raised its countercyclical capital buffer (CCyB) to 2% while the rulebook said 0%. It was refilling a buffer, not fighting a boom

SUMMARY

De Nederlandsche Bank (DNB) raised the CCyB from 0% to 1% (announced May 2022) and then to 2% (announced May 2023, binding since 31 May 2024). At both decisions the Basel reference indicator, the credit-to-GDP gap, stood far below zero (−33 and −47 percentage points), so the rulebook pointed to a buffer of 0%.

Q1 · ENVIRONMENT

The economy had recovered from COVID, but war, an energy shock and 11.6% inflation darkened the outlook, and the ECB began raising rates in July 2022. Credit was cold, housing was hot and then cooled, and the banks were strong and growing more profitable.

Q2 · CRITERIA AND RISKS

DNB's 2022 framework sets 2% as the normal level, calibrated on past crisis losses rather than on credit growth. The buffer insures against shocks that no indicator predicts, while housing risks are handled by other tools. It also kept a promise made in March 2020.

Q3 · IMPLICATIONS

The buffer was cheap, at most about a fifth of the big banks' spare capital. It was largely a swap for cuts in structural buffers, and we see no sign of a contraction in lending. The Netherlands now has €6.7bn it can release.

Our verdict: **normalisation, not tightening**. The price is a greater reliance on DNB's judgement, and the buffer has not yet been tested in a crisis.

Source: DNB (2020, 2022, 2023, 2026); DNB Financial Stability Reports 2022–23; BIS credit-to-GDP gaps; ECB; bank disclosures; own calculations.
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STORY · what to say

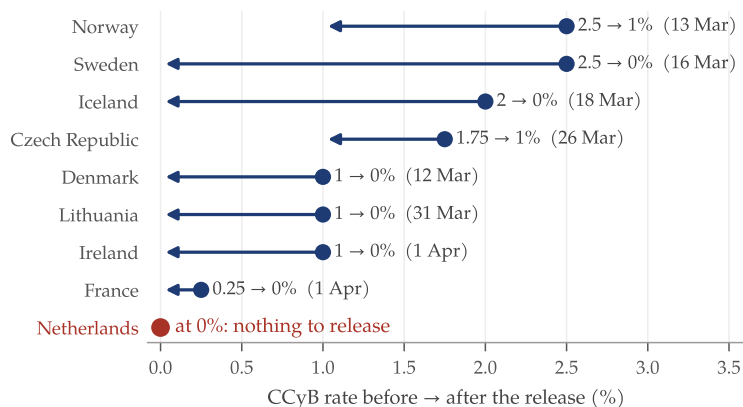
- Here is the whole story on one slide. DNB raised the buffer in two steps, to one percent and then to two percent. Yet at both decisions the indicator at the centre of the Basel rules, the credit-to-GDP gap, was deeply negative, so the rulebook said the buffer should be zero.
- The three columns follow the three questions of the assignment: the environment, the criteria and risks, and the implications.
- Our verdict is that this was **normalisation, not tightening**. DNB was refilling an empty buffer, not fighting a credit boom. The price is that the policy relies on DNB's judgement rather than on a rule.

LIKELY CHALLENGES · definitions and logic

- Q Define the credit-to-GDP gap.
 - Credit to the private non-financial sector as a share of GDP, minus its long-run trend from a one-sided HP filter ($\lambda = 400,000$). It is the Basel starting point for setting the CCyB.
- Q What is the Basel buffer guide?
 - A mapping from the gap to a buffer rate: below 2pp → 0%; above 10pp → 2.5%; linear in between. With gaps of −33 and −47pp it gives 0%.
- Q What do you mean by 'normalisation, not tightening'?
 - Normalisation means returning the buffer to its planned normal-times level regardless of the cycle. Tightening means raising it because risk is building up. We test which one fits the evidence on page 14.
- Q Isn't raising capital always a tightening?
 - Not if other requirements are cut at the same time. The 2020 and 2024 cuts in structural buffers roughly offset the CCyB relative to pre-COVID (page 15).

In March 2020 COVID struck and neighbours released their buffers, but the Dutch buffer was empty

PROLOGUE



Within three weeks, eight European countries cut their countercyclical buffers so that banks could absorb losses and keep lending. Belgium and Germany cancelled increases they had already announced.

The Netherlands had nothing to release. DNB improvised: on 17 March 2020 it cut the systemic buffers of ING, Rabobank and ABN AMRO and postponed a planned floor on mortgage risk weights. This freed €8bn of capital, enough to support up to €200bn of lending.

“Once the situation is back to normal, DNB will compensate the systemic buffers reduction by gradually increasing the countercyclical capital buffer to 2%.” DNB, 17 March 2020

Source: ESRB, CCyB rates table (announcements March–April 2020); DNB press release “DNB lowers bank buffer requirements to support lending”, 17 Mar 2020.
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STORY · what to say

- The story starts in March 2020. When COVID hit, our neighbours reacted within three weeks: eight European countries cut their countercyclical buffers, as the chart shows. Sweden and Norway had been at two and a half percent; Denmark, Ireland and Lithuania at one percent. Buffers built in good times were used exactly as intended.
- The Netherlands was at **zero**, the red dot at the bottom. There was nothing to release.
- So DNB improvised. It cut the systemic buffers of ING, Rabobank and ABN AMRO and postponed a planned floor on mortgage risk weights. This freed about eight billion euros of capital, enough to support up to 200 billion euros of lending. But those buffers were never designed to be released.
- And DNB made a promise (*pause*): once things were back to normal, it would rebuild this capital as a two-percent countercyclical buffer.

LIKELY CHALLENGES · definitions and logic

Q What does ‘release’ mean legally?

- The authority lowers the CCyB rate with immediate effect. The requirement itself falls, so banks can use that capital to absorb losses or lend without breaching any requirement.

Q If the Dutch CCyB was 0%, what did DNB release in 2020?

- It cut the systemic risk buffers of ING, Rabobank and ABN AMRO (3% → 2.5/2/1.5%) and postponed the mortgage risk-weight floor. Together about €8bn, supporting up to €200bn of lending.

Q Why was that a ‘workaround’?

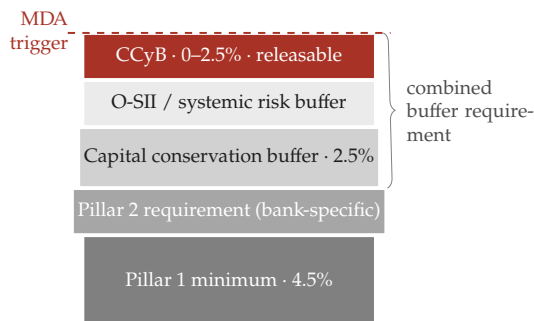
- Systemic buffers are structural charges for the size and importance of banks. They are not meant to move with the cycle, so cutting them blurred their purpose. That is why DNB promised to restore the capital as a releasable 2% CCyB.

Q Which countries released, and by how much?

- Norway 2.5→1%, Sweden 2.5→0%, Iceland 2→0%, Czech Republic 1.75→1%, Denmark, Lithuania and Ireland 1→0%, France 0.25→0%. Belgium and Germany cancelled planned increases (ESRB table).

That mattered because the CCyB is the only layer of bank capital an authority can release in a crisis

PROLOGUE



Capital stack in % of risk-weighted assets (RWA), not to scale.

Read the stack from the bottom. Banks must always meet the minimum requirements; the buffers on top can be used in bad times.

If a bank's capital falls below the top of its buffers (the MDA trigger), its dividends and bonuses are restricted automatically. Banks therefore avoid using buffers even when they are allowed to. In COVID, banks close to this line cut lending instead.

The CCyB works differently. The authority can switch it off, so the requirement itself falls and the bank faces no penalty and no stigma. Increases bind only after twelve months, but a release is immediate.

Only a buffer that the authority can **release** is truly usable in a crisis, and it has to be filled beforehand.

Source: CRD (Arts. 130–136); BCBS (2024) d585; Couaillier, Lo Duca, Reghezza & Rodriguez d'Acri (2022), ECB WP 2644.
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STORY · what to say

- Why did an empty buffer matter? Because not all bank capital can be used in a crisis.
- Read the stack from the bottom: first the minimum requirements, then the buffers on top. The red dashed line is the MDA trigger. If a bank falls below it, its dividends and bonuses are restricted automatically, and markets read that as weakness. So in COVID, banks close to this line cut lending rather than use buffers they were formally allowed to use.
- The countercyclical buffer is different. The authority can switch it off, so the requirement itself falls, with no penalty and no stigma. It is the only truly **releasable** layer. And while an increase binds only after twelve months, a release is immediate, so the buffer must be filled before a crisis.

LIKELY CHALLENGES · definitions and logic

- Q Define CET1 and RWA.**
 - CET1 (Common Equity Tier 1) is the highest-quality capital: shares and retained earnings. RWA (risk-weighted assets) weight each exposure by its riskiness, so capital ratios are CET1 / RWA.
- Q Walk through the capital stack.**
 - Pillar 1 minimum 4.5% CET1, then the bank-specific Pillar 2 requirement, then the combined buffer requirement: capital conservation buffer 2.5%, O-SII/systemic buffer and CCyB. Pillar 2 guidance sits on top and is not binding.
- Q What is the MDA trigger?**
 - The maximum distributable amount. If CET1 falls below the top of the combined buffer requirement, dividends, buybacks and bonuses are automatically restricted.
- Q Buffers are 'usable' anyway; why do banks not use them?**
 - Falling below the MDA trigger means payout limits, market stigma and rating pressure. COVID evidence shows banks close to the trigger cut lending instead (Couaillier et al., 2022). A released CCyB avoids this because the requirement itself falls.
- Q Why is the 12-month lag important?**
 - An increase binds only 12 months after the announcement, but a release is immediate. So the buffer must be filled before a crisis; it cannot be built once the shock arrives.

Two years later DNB refilled it in the middle of a storm and against its own rulebook. Was that the wrong moment?

THE QUESTION

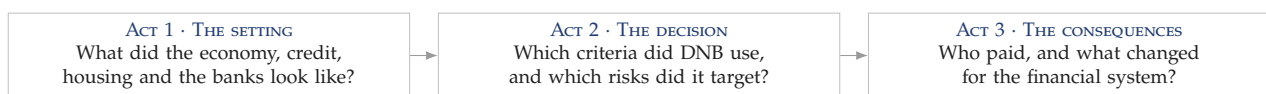
THE STORM OF 2022–23

Russia invaded Ukraine and energy prices hit records. Dutch inflation averaged 11.6% in 2022, the ECB raised rates from July 2022, and house prices were about to fall. The Basel rule, based on the credit gap, said the buffer should be 0%.

THE DECISION

DNB nevertheless raised the CCyB from 0% to 1% (announced May 2022) and then to 2% (announced May 2023). Each step became binding a year later, in May 2023 and May 2024.

Raising capital requirements in a downturn can amplify the cycle, which is exactly the **procyclical mistake** the CCyB was invented to avoid. So was DNB tightening into a slowdown, fighting a hidden boom, or doing something else?



Source: CBS (HICP 2022); ECB monetary policy decision, 21 Jul 2022; BIS credit-to-GDP gaps (latest data at each decision: 2021Q4, 2022Q4).
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STORY · what to say

- Two years later, in 2022, DNB began to refill the buffer, and the timing looks strange. On the left is the storm: war, record energy prices, 11.6 percent inflation, the first ECB rate hikes and a housing market about to turn. And the Basel indicator said zero. On the right is the decision: one percent, then two percent.
- Raising capital requirements in a downturn can amplify the cycle. That is exactly the procyclical mistake the buffer was invented to avoid.
- So our question is: **was this the wrong moment?** Was DNB tightening into a slowdown, fighting a hidden boom, or doing something else? We answer in three acts, one for each question of the assignment.

LIKELY CHALLENGES · definitions and logic

Q What is procyclicality?

- Policy or behaviour that reinforces the cycle: tighter in bad times, looser in good times. Raising capital in a downturn can force banks to cut lending exactly when the economy needs it.

Q Why was this moment suspicious?

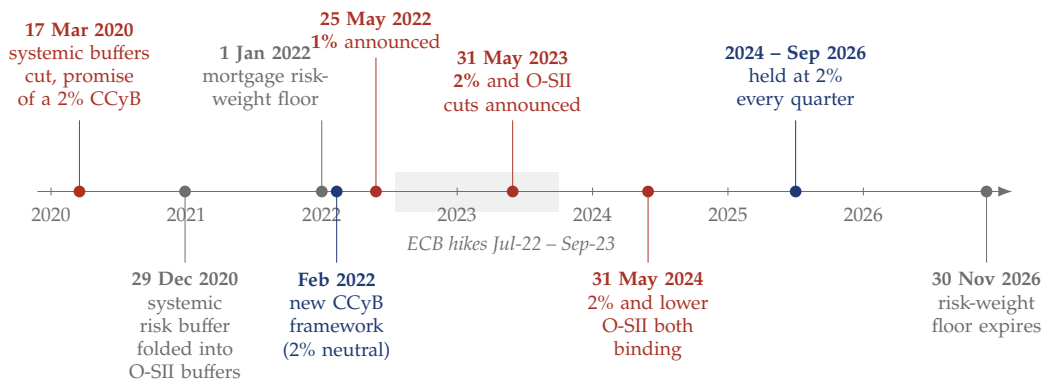
- War, 11.6% inflation (2022), ECB hikes from July 2022 and a turning housing market, while the Basel indicator said 0%. On the surface it looks like the procyclical mistake the CCyB was designed to avoid.

Q How do the three acts map to the assignment?

- Act 1 = economic and financial environment;
Act 2 = criteria and financial-stability risks;
Act 3 = wide-ranging implications.

The refill followed the path promised in 2020, one step per year, just as the ECB started to tighten

Q1 · ENVIRONMENT



DNB moved in two equal steps of one percentage point, a year apart, and then stopped at 2%. The first step was announced two months before the ECB's first rate hike.

Source: DNB press releases (17 Mar 2020; 27 May 2022; 31 May 2023; 24 Apr 2026; 17 Sep 2026); ESRB O-SII notifications (27 Nov 2020; 31 May 2023); ECB (21 Jul 2022; 14 Sep 2023).

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STORY · what to say

- Act one is the setting, and it starts with the timeline. In March 2020 DNB cut the systemic buffers and made its promise. In February 2022 it published a framework with two percent as the normal level. In May 2022 it announced one percent, two months before the ECB's first hike. In May 2023 it announced two percent, together with lower buffers for the largest banks, and both became binding in May 2024.
- Since then DNB has held two percent at every quarterly review. So the path was two equal steps, a year apart, and then a stop. Keep that pattern in mind for our verdict.

LIKELY CHALLENGES · definitions and logic

Q Announcement date vs binding date?

- Banks get 12 months between an increase being announced and having to meet it. So 1% was announced in May 2022 and binding in May 2023; 2% announced in May 2023 and binding in May 2024.

Q Was the increase a response to ECB tightening?

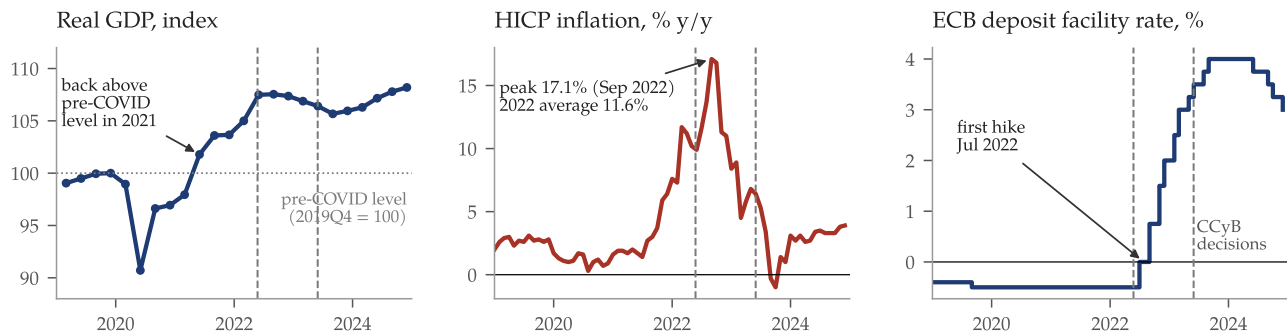
- No. The 1% was announced on 25 May 2022, two months before the ECB's first hike on 21 July 2022, and the path had been promised in 2020.

Q Why did the systemic risk buffer become an O-SII buffer in December 2020?

- CRD V made the O-SII and systemic risk buffers additive, so DNB converted the large banks' systemic risk buffers into O-SII buffers from 29 December 2020.

By then the economy had fully recovered, but war, energy prices and inflation were darkening the outlook

Q1 · ENVIRONMENT



GDP was back above its pre-COVID level in 2021, which is the condition DNB's framework sets before a buffer is built. Then inflation rose to a peak of 17.1%, and the ECB moved its deposit rate from -0.5% to 4% in fourteen months. DNB wrote in spring 2022 that *"the economic outlook has worsened due to the war in Ukraine and high inflation."*

The macro picture argued both ways: build the buffer while the economy can take it, or wait because it is weakening.

Source: ECB Data Portal (Dutch real GDP, MNA; Dutch HICP, ICP; deposit facility rate, FM), current vintage; DNB Financial Stability Report, spring 2022. Dashed lines: CCyB announcements.

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STORY · what to say

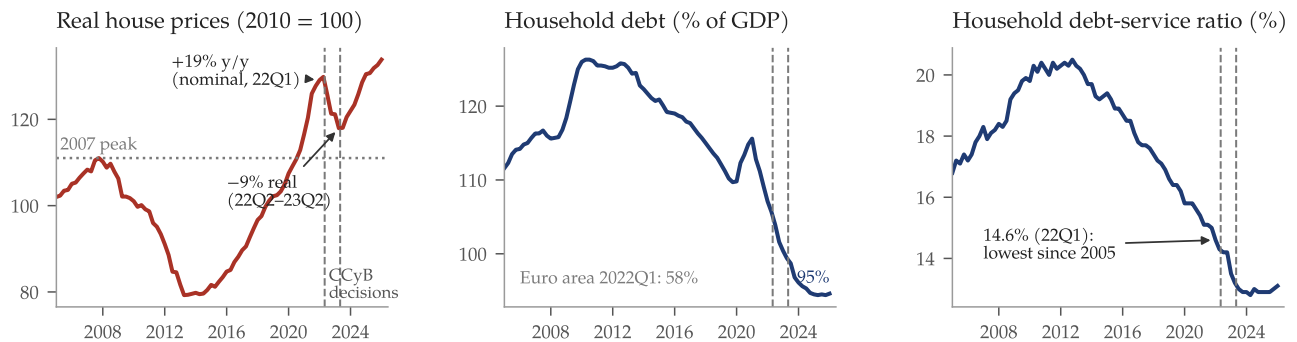
- What did the economy look like? The left panel shows that real GDP was back above its pre-COVID level during 2021, which is the condition DNB's framework sets before rebuilding the buffer.
- Then the storm arrived. Inflation, in the middle, peaked at 17 percent in September 2022, and on the right the ECB moved its deposit rate from minus half a percent to four percent in fourteen months. DNB warned that higher funding costs could strain the debt of governments, firms and households.
- So the macro picture argued **both ways**: build while the economy can take it, or wait because it is weakening.

LIKELY CHALLENGES · definitions and logic

- Q **Why does the recovery matter for the decision?**
 - DNB's framework only builds the buffer once the economy has recovered from a crisis (phase 1 → phase 2). GDP was back above its pre-COVID level during 2021.
- Q **Which inflation measure do you use?**
 - HICP, the harmonised index used across the euro area: 11.6% average in 2022 and a peak of 17.1% in September 2022. The national CPI was 10.0% in 2022.
- Q **Why GDP 'in 2021' rather than a quarter?**
 - In current data GDP is back above 2019Q4 in 2021Q2; the first CBS estimate dated it to 2021Q3. Data revisions explain the difference, so we say 2021.
- Q **Define a basis point.**
 - 0.01 percentage point. The ECB's first hike was +50bp; the deposit rate went from -0.5% to 4% in fourteen months.

In the financial system credit was cold and only housing was hot, and even housing soon turned

Q1 · ENVIRONMENT



Housing was hot. Nominal prices rose 19% in the year to early 2022, and DNB itself wrote that “*the overheated housing market calls for further measures.*” As mortgage rates rose, real prices fell by 9%.

Credit was not. Household debt is high (107% of GDP against 58% in the euro area) but it was falling, and the share of income spent on debt service was the lowest since 2005.

If there was a boom, it was in house prices, not in credit.

Source: BIS (WS_SPP, WS_TC, WS_DSR); DNB Financial Stability Report, spring 2022; own calculations. Dashed lines: CCyB announcements (May 2022, May 2023). Huang & Gutiérrez · The Netherlands' CCyB

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STORY · what to say

- What about the financial system? On the left you see housing, and housing was hot. Nominal house prices rose 19 percent in a single year, real prices were well above their 2007 peak, and DNB itself called the market overheated. But as mortgage rates rose, real prices fell by about nine percent.
- In the middle is household debt. It is high, at 107 percent of GDP against 58 percent in the euro area, but it was **falling**. On the right is the share of income that households spend on interest and repayments. It was the lowest since 2005.
- So if there was a boom, it was in house prices, not in credit.

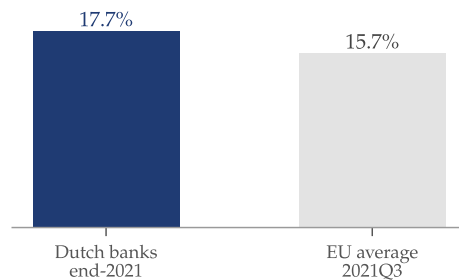
LIKELY CHALLENGES · definitions and logic

- Q **Define the debt-service ratio.**
 - Interest plus principal payments as a share of household income. At 14.6% in 2022Q1 it was the lowest since 2005.
- Q **Real vs nominal house prices?**
 - Real prices are nominal prices deflated by consumer prices. Nominal prices rose 19% y/y in 2022Q1, but real prices fell about 9% between 2022Q2 and 2023Q2.
- Q **Household debt is 107% of GDP. Isn't that a risk in itself?**
 - Yes, it is a structural vulnerability, which is why the buffer exists. But it was falling, so it is not a sign of a cyclical credit boom.
- Q **Didn't debt/GDP fall only because inflation raised nominal GDP?**
 - Partly. Debt/GDP fell from 110% (2019Q4) to 100% (2023Q1) while the debt stock rose 12%, because nominal GDP grew about 22%. That is a denominator effect, and another reason not to read ratios mechanically.

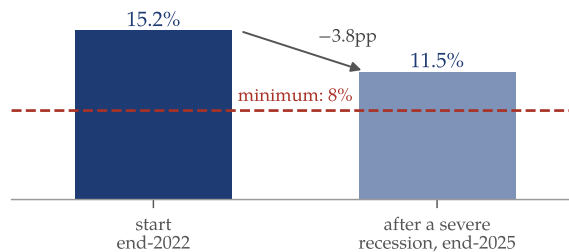
Banks entered this storm well capitalised, and rising rates were lifting their profits

Q1 · ENVIRONMENT

Core capital (CET1) ratio, % of RWA



DNB stress test, four largest banks' CET1 ratio



Dutch banks held more core capital than the EU average, and in DNB's stress test the four largest banks stayed well above the minimum after a severe recession. Rising rates also lifted their net interest income, by 6.7% in 2022; DNB had expected higher rates to *"ultimately have a positive impact on the profitability of financial institutions."*

Act 1 in one sentence: war, inflation, rate hikes and a turning housing market argued against raising the buffer, while a recovered economy, cold credit and strong banks argued for it.

Source: DNB Financial Stability Reports, spring 2022 (CET1 17.7%, EU 15.7%) and spring 2023 (stress test, net interest income).
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STORY · what to say

- Finally, the banks. Dutch banks entered the storm with a core capital ratio of 17.7 percent, two points above the EU average. In DNB's 2023 stress test, the four largest banks fell by 3.8 points in a severe recession but stayed at 11.5 percent, well above the minimum.
- Rising rates also **lifted their profits**: net interest income grew 6.7 percent in 2022. Please remember this, because it will come back in act three.
- So act one in one sentence: war, inflation, rate hikes and a turning housing market argued against raising the buffer, while a recovered economy, cold credit and strong banks argued for it.

LIKELY CHALLENGES · definitions and logic

Q What does a stress test show?

- A simulation of a severe recession to check whether capital stays above the minimum. In DNB's 2023 test the four largest banks fell by 3.8pp, from 15.2% to 11.5%, still above the 8% minimum used.

Q Define net interest income.

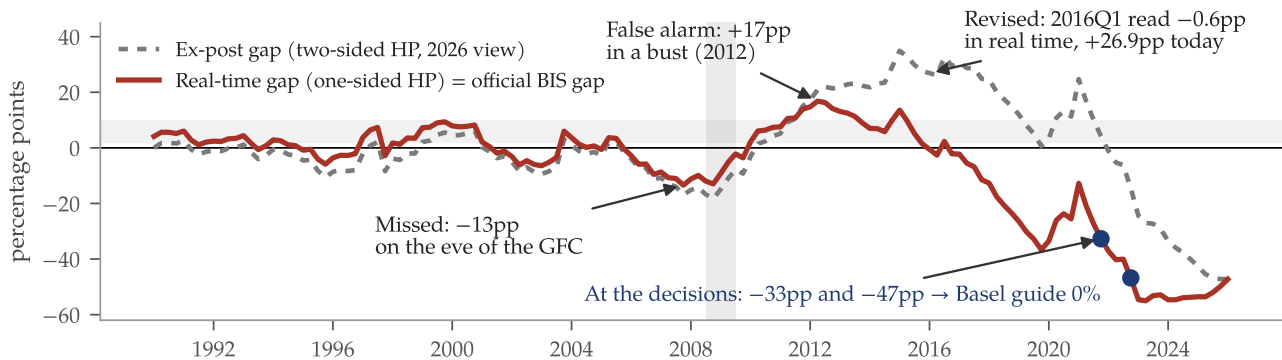
- Interest earned on loans minus interest paid on deposits and funding. It rose 6.7% in 2022 and makes up 67.7% of Dutch banks' income.

Q Why does profitability matter for the CCyB?

- Banks can meet a higher requirement from retained earnings instead of cutting assets. Higher rates raised profits, which made the build-up cheaper (page 16).

So why ignore the rulebook? On Dutch data the Basel rule had missed the last crisis and raised false alarms

Q2 · CRITERIA & RISKS



The credit-to-GDP gap measures how far credit relative to GDP is above its long-run trend. Under the Basel guide the buffer starts when the gap exceeds 2 points and reaches 2.5% at 10 points.

On Dutch data the gap missed the 2008 crisis, sounded a false alarm in 2012 and was later revised. At the two decisions it read -33 and -47 points.

Source: BIS credit-to-GDP gaps (Q.NL.P.A); Drehmann et al. (2010); Edge & Meisenzahl (2011); own one-sided HP filter ($\lambda = 400,000$).
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STORY · what to say

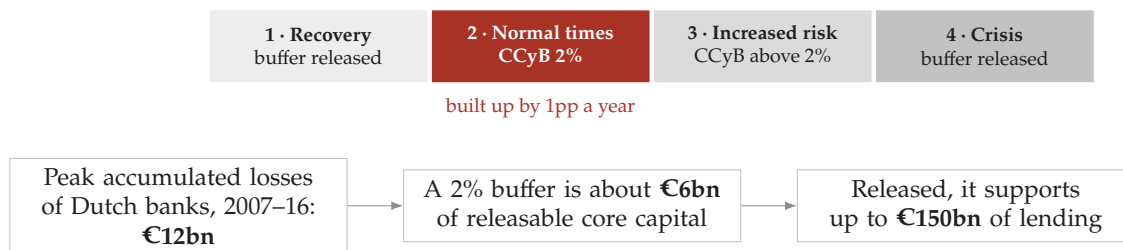
- Act two is the decision, and the first question is: why did DNB ignore the rulebook? Because on Dutch data the rulebook has a poor record.
- The red line is the credit-to-GDP gap: how far credit relative to GDP is above its long-run trend. Under the Basel guide the buffer starts when the gap exceeds two points. We rebuilt the gap ourselves, and it matches the official BIS series almost exactly.
- Look at its record. It **missed** the financial crisis, reading minus 13 on the eve of 2008. It raised a **false alarm** in 2012, in the middle of a bust. And it gets **revised**: 2016 read slightly negative at the time and plus 27 with today's data. The reason is that Dutch credit grew from about 50 to 350 percent of GDP, and the filter mistakes that structural rise for a trend.
- At the two decisions the gap read minus 33 and minus 47. Taken literally, the rule said zero.

LIKELY CHALLENGES · definitions and logic

- Q How does the HP filter work, and why $\lambda = 400,000$?**
 - It finds a smooth trend that balances fit against smoothness. $\lambda = 1,600$ is standard for business cycles; credit cycles are about four times longer, and λ scales with the fourth power of that ratio: $1,600 \times 4^4 \approx 400,000$.
- Q One-sided vs two-sided filter?**
 - One-sided uses only data up to each date, which is what policymakers see; it is the official BIS/Basel gap. Two-sided also uses later data, so it is hindsight. 2016Q1: -0.6pp one-sided vs +26.9pp two-sided.
- Q Type I vs type II error?**
 - A missed crisis (no signal before a crisis, e.g. -13pp before 2008) vs a false alarm (signal without a crisis, e.g. +17pp in 2012).
- Q Why does the gap fail especially in the Netherlands?**
 - Credit/GDP rose from about 47% (1961) to 352% (2015) for structural reasons such as a large, tax-favoured mortgage market. The filter treats this as trend and then produces large negative gaps when the ratio falls.
- Q Then why does Basel still use it?**
 - Across many countries it is one of the better early-warning indicators. Basel treats it as a starting point, not a mechanical trigger.

DNB used its own compass instead: 2% in normal times, calibrated on past crisis losses rather than on credit growth

Q2 · CRITERIA & RISKS



The framework sets the buffer at 2% when risks are neither high nor low, a so-called *positive neutral* buffer. DNB reads a dashboard of indicators but stresses that there is “no mechanical link . . . guided discretion.” Seventeen jurisdictions now work this way, and Sweden, the UK and Poland also target 2%.

The 2% is sized to absorb about half of past crisis losses. It does not depend on credit growth.

Source: DNB (2022), Analytical framework for setting the CCyB in the Netherlands; BCBS (2024) d585, Table 1.
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STORY · what to say

- So DNB used its own compass. Its 2022 framework has four phases: release after a crisis, **two percent in normal times**, more than two percent when risks are clearly elevated, and release again in a crisis. This is a positive neutral buffer: it is already filled when risks are neither high nor low.
- The two percent comes from history, not from credit growth. At the peak, past crises cost Dutch banks about 12 billion euros. Two percent is about six billion euros of releasable capital, roughly half of those losses, and enough to support up to 150 billion euros of lending. The ECB's own estimates range from 1.1 to 1.8 percent, so two percent is at the top of the range, but not outside it.
- Seventeen jurisdictions now work this way, and Sweden, the UK and Poland also target two percent.

LIKELY CHALLENGES · definitions and logic

- Q Define a positive neutral CCyB.**
 - A CCyB kept above zero when risks are neither high nor low, so there is always something to release, whatever the source of the shock. BCBS (2024) lists 17 jurisdictions using one.
- Q What model did DNB use for 2%?**
 - No econometric model. It sized the buffer on history: peak accumulated losses of Dutch banks in 2007–16 were about €12bn; 2% ≈ €6bn of releasable capital, about half of those losses, supporting up to €150bn of lending.
- Q Define peak accumulated losses.**
 - The largest cumulative loss over a crisis period, from the start of losses to their worst point.
- Q What is guided discretion?**
 - A published framework (phases, neutral rate, indicator dashboard) plus judgement. DNB: there is ‘no mechanical link between the indicator values and the level of the CCyB’.
- Q Is 2% too high?**
 - The ECB's loss-based estimates give 1.1–1.8% for the euro area, so 2% is at the top. DNB also took the 2020 buffer cut into account. We list this as a critique on page 21.

Its stated reasons were recovery, uncertainty and the 2020 promise, never credit growth

Q2 · CRITERIA & RISKS

MAY 2022 · FROM 0% TO 1%

*"When the systemic risk buffers . . . were lowered in March 2020, we also announced our intention to **restore the buffers** by raising the CCyB."*

"The risk profile is currently dominated by uncertainty caused by the war in Ukraine, but . . . the robust economic recovery . . . provides grounds for a gradual build-up."

DNB judged cyclical risks to be normal to elevated.

MAY 2023 · FROM 1% TO 2%

*"Although the credit-to-GDP gap shows **no signs of excessive credit growth**, some other indicators already point to higher cyclical risk."*

DNB cited investors' rising appetite for risk, falling real-estate prices, the debt sustainability of firms and governments, and banks that were "robust, partly due to rising interest rates."

Both decisions rested on the same three pillars: a normal-to-elevated risk picture, strong banks and the 2020 promise. Credit growth was never the reason.

Source: DNB Financial Stability Report, spring 2022; DNB press releases of 27 May 2022 and 31 May 2023.
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STORY · what to say

- What reasons did DNB actually give? In May 2022 it wrote that it was **restoring the buffers** it had lowered in March 2020. It acknowledged the uncertainty from the war, but said the robust recovery justified a gradual build-up.
- In May 2023 it went further. It wrote that the credit gap showed **no signs of excessive credit growth**. Instead it pointed to investors' rising risk appetite, falling property prices, the debt sustainability of firms and governments, and banks that were robust, partly thanks to higher rates.
- So both decisions rested on the same three pillars: a normal-to-elevated risk picture, strong banks and the 2020 promise. Credit growth was never the reason.

LIKELY CHALLENGES · definitions and logic

Q Define cyclical systemic risk.

- Risk that builds up over the financial cycle (credit, asset prices, risk-taking) and can materialise system-wide, as opposed to structural risk from the size or concentration of the banking sector.

Q Did DNB ever cite credit growth?

- No. In 2023 it wrote that the gap showed 'no signs of excessive credit growth' and pointed to risk appetite, falling property prices, debt sustainability and strong banks.

Q Isn't 'uncertainty' an argument to wait?

- It cuts both ways. Uncertainty is exactly why a releasable buffer is valuable; DNB judged the recovery strong enough to build it gradually.

The buffer was meant for shocks nobody can forecast, while the known hot spots got their own tools

Q2 · CRITERIA & RISKS

Shocks nobody can forecast

war, energy, a sudden tightening of financial conditions, bank turmoil abroad (SVB, Credit Suisse)

CCyB

releasable whatever the source of the shock

The known hot spot: housing

an “overheated” market, high mortgage debt of first-time buyers, interest-only loans

Borrower limits, tax reform, mortgage risk-weight floor

(floor in force Jan 2022 – Nov 2026)

The size of the big banks

a banking sector of about 280% of GDP (end-2022)

O-SII buffer

structural, for systemically important banks

One instrument per target, as the Tinbergen principle suggests. If housing had been the target, the CCyB would be the wrong tool, because it raises the cost of all lending, not only mortgages.

Source: DNB Financial Stability Reports, spring 2022 and spring 2023; ESRB O-SII notification NL (31 May 2023); DNB (24 Apr 2026); Tinbergen (1952).
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STORY · what to say

- Which risks was the buffer meant to cover? DNB matched each risk with its own tool.
- The top row is **shocks nobody can forecast**: war, an energy shock, a sudden tightening of financial conditions, or bank turmoil abroad like Silicon Valley Bank and Credit Suisse in 2023. These go to the countercyclical buffer, because it can be released whatever the source of the shock.
- **Housing**, the known hot spot, was handled by borrower limits, tax reform and a mortgage risk-weight floor. The **size of the big banks** is handled by the structural buffer for systemically important banks.
- This is the Tinbergen principle: one instrument per target. If housing had been the target, the countercyclical buffer would have been the wrong tool, because it raises the cost of all lending.

LIKELY CHALLENGES · definitions and logic

- Q **State the Tinbergen principle.**
 - To reach n independent policy targets you need at least n instruments, and each instrument should go to the target it affects most directly (Tinbergen, 1952).
- Q **Define LTV and LTI.**
 - Loan-to-value and loan-to-income limits: caps on the size of a mortgage relative to the house value or the borrower's income. They are borrower-based measures.
- Q **What does the mortgage risk-weight floor do?**
 - It sets a minimum average risk weight on Dutch mortgages for banks using internal models (Art. 458 CRR), raising RWA and capital for housing risk. In force January 2022 to November 2026.
- Q **Why not use the CCyB against housing?**
 - It raises the cost of all lending, not just mortgages, so it is blunt for a sectoral problem and weak at steering house prices. Its comparative advantage is resilience against any shock.
- Q **Define O-SII.**
 - Other systemically important institution. The O-SII buffer is a structural charge for size and interconnectedness, not designed to be released.

Tested against the evidence, every prediction fits normalisation and none fits overheating

Q2 · CRITERIA & RISKS

	If DNB was normalising	If DNB was fighting a boom	What actually happened	
Credit indicators	the buffer ignores them	the buffer rises with them	the indicators fell while the buffer rose	✓
The path	+1pp a year, stop at 2%	follows the risk	1%, then 2%, then held	✓
2023: prices falling, rates jumping	raise anyway	pause	DNB raised	✓
2024–26: house prices up 10% again	stay at 2%	go above 2%	held at 2%	✓
Other buffers	cut structural buffers in return	keep them	cut in 2020 and 2024	✓

Five predictions, five matches for **normalisation**. The last row is the key to the next slide.

Source: BIS (WS_CREDIT_GAP, WS_DSR, WS_TC, WS_SPP); DNB press releases 2022–2026 (house prices: DNB, 26 Mar 2026); DNB framework (2022); ESRB O-SII notifications. Charts for each row: Appendix A1–A2.
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STORY · what to say

- Now we can test the two readings. If DNB was fighting a boom, the buffer should follow credit and prices. If it was normalising, it should follow a fixed path back to its normal level. Row by row:
- **Credit indicators:** they all fell while the buffer rose. **The path:** one point a year and a stop at two, exactly as normalisation predicts. **2023:** prices falling and rates jumping; a boom fighter would pause, but DNB raised. **2024 to 2026:** house prices up about ten percent again; a boom fighter would go above two percent, but DNB held. **Other buffers:** normalisation means cutting structural buffers in return, and DNB did so twice.
- *(pause)* Five predictions, five matches for normalisation, and none for overheating. This is the core of our argument.

LIKELY CHALLENGES · definitions and logic

Q Why is this a valid test?

- The two hypotheses make different, observable predictions about the path, the timing and the other buffers. Five out of five fit normalisation and none fits overheating. Charts for each row: Appendix A1–A2.

Q What is framework phase 3?

- 'Increased risk': the phase in which DNB would set the CCyB above 2%. It has not gone there, even when house prices rose about 10% a year in 2024–25.

Q Couldn't DNB be fighting a boom it expected later?

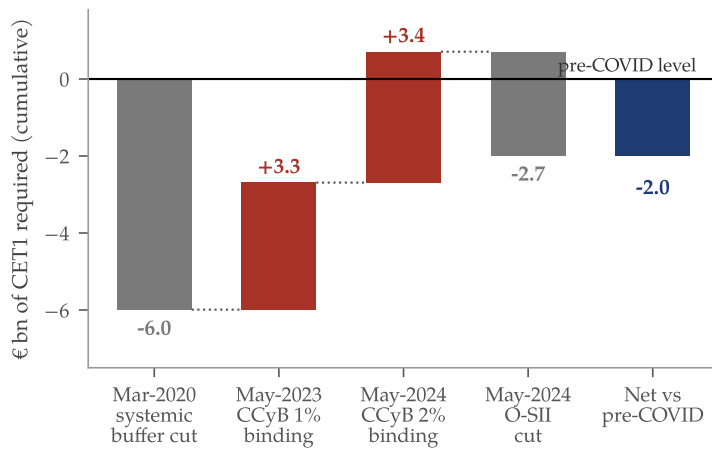
- Then it would have raised further when house prices rebounded in 2024–25. It held at 2%.

Q Is this causal evidence?

- No. It is a consistency test of DNB's behaviour against two explanations, not an estimate of effects.

And the refill was a swap, not a squeeze: capital moved from structural buffers into the releasable one

Q2 · CRITERIA & RISKS



By our estimate, the 2020 cut released about €6bn and the 2024 cut another €2.7bn, while the two CCyB steps added €3.3bn and €3.4bn. Measured against the pre-COVID requirement, banks hold roughly the same amount of capital, or slightly less.

DNB described the swap as “more or less capital-neutral for the three large banks” (2020) and later as “a limited increase in the net capital requirements” (Knot, 2023).

Capital moved from buffers that cannot be released into one that can. Relative to 2021, however, it was a modest increase.

Source: Own calculation (cuts valued at end-2022 RWA of the four largest banks; CCyB amounts: DNB, sector-wide); DNB (17 Mar 2020); Knot (7 Jun 2023).
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STORY · what to say

- And the last row shows that the refill was a swap, not a squeeze. By our estimate, the 2020 cut released about six billion euros, the two countercyclical steps added 3.3 and 3.4 billion, and the 2024 cut released another 2.7 billion. Net, compared with before COVID, banks hold roughly the same required capital, or slightly less. DNB called it more or less capital-neutral for the three large banks.
- Two caveats: the bases differ, so the numbers are approximate, and compared with 2021 the refill was a modest increase.
- So capital was **moved** from buffers that cannot be released into one that can. The total stayed similar; what changed is who can use it, and when.

LIKELY CHALLENGES · definitions and logic

Q Define a structural buffer.

- A capital charge for permanent features of a bank, such as size and interconnectedness (O-SII, systemic risk buffer). It is kept through the cycle and not meant to be released.

Q How did you compute the swap?

- Own estimate: buffer cuts valued at end-2022 RWA of the four largest banks (–€6.0bn in 2020, –€2.7bn in 2024), plus DNB's sector-wide CCyB amounts (+€3.3bn, +€3.4bn). Net about –€2.0bn vs pre-COVID.

Q What are the caveats?

- Different bases: O-SII applies to consolidated RWA, the CCyB only to Dutch exposures; the CCyB amounts include foreign and small banks; BNG is excluded. Relative to 2021 the refill was a modest increase.

Q Why not keep the structural buffers and add the CCyB?

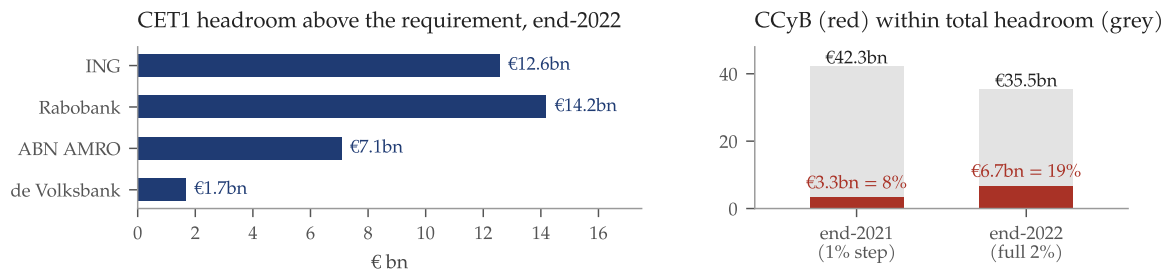
- The goal was usability, not quantity. Releasable capital is worth more per euro in a crisis, and stacking both would have been a real tightening.

Q Did other countries do this?

- Yes. The UK offset its 2% neutral CCyB by lowering Pillar 2A and resolution requirements.

So was it the wrong moment? No. Higher rates gave banks the profits to build the buffer cheaply

Q3 · IMPLICATIONS



Headroom is the core capital a bank holds above its requirement. The four largest banks had €35.5bn of it at the end of 2022, so the full buffer of €6.7bn used at most a fifth. Higher net interest income meant the buffer could be built from retained earnings rather than by cutting loans, and the ECB finds that a gradual build-up by profitable banks keeps the costs low.

It was not the wrong moment. Arguably it was the right one.

Source: Bank disclosures (ING, Rabobank, ABN AMRO, de Volksbank, 2021–22), de Volksbank requirement derived; DNB FSR 2023; ECB Macroprudential Bulletin 24 (2024) and 31 (2025, Box 2).
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STORY · what to say

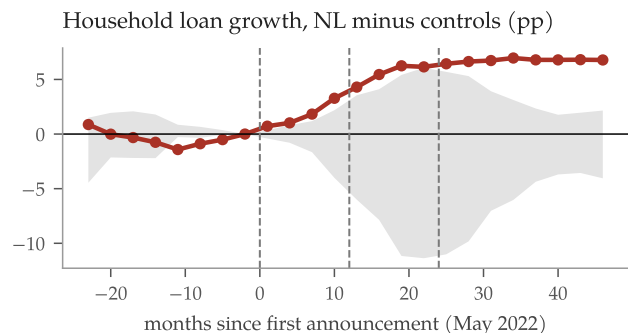
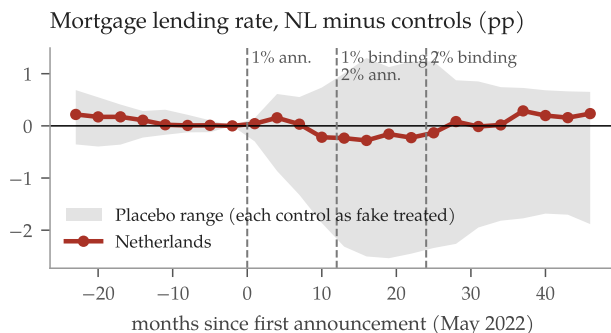
- So, was it the wrong moment? We think not. The storm actually made it the **cheapest** moment.
- Headroom is the capital a bank holds above its requirement. At the end of 2022 Rabobank had 14.2 billion euros, ING 12.6, ABN AMRO 7.1 and de Volksbank 1.7, about 36 billion in total. The first step of 3.3 billion used eight percent of that, and the full buffer of 6.7 billion at most a fifth.
- And here is the twist. The same rising rates that made the moment look dangerous lifted bank profits, so banks could build the buffer from retained earnings instead of cutting loans. ECB research agrees that profitable banks and a gradual build-up keep the costs low.

LIKELY CHALLENGES · definitions and logic

- Q Define headroom.**
 - CET1 capital above a bank's requirement, i.e. above its MDA trigger. Headroom = (CET1 ratio – requirement) × RWA.
- Q Where do the bank numbers come from?**
 - Bank disclosures for ING, Rabobank and ABN AMRO. Only de Volksbank's requirement is derived (4.5% + 1.41% P2R + 2.5% + 1.0% O-SII).
- Q Why is 19% an upper bound?**
 - We compare DNB's sector-wide CCyB amount (all banks, Dutch exposures) with the headroom of only four banks, so the true share is lower.
- Q Doesn't Modigliani–Miller say capital is free?**
 - In frictionless markets more equity lowers the required return on equity, so funding costs barely change. Taxes, implicit guarantees and issuance costs make it costly in practice, but the cost is small when banks can retain earnings.
- Q Did headroom fall because of the CCyB?**
 - No. It fell from €42.3bn to €35.5bn mainly through RWA growth and payouts; the first CCyB step only bound in May 2023.

Borrowers did not pay either: credit kept flowing and its price moved with the rest of the euro area

Q3 · IMPLICATIONS



We compare the Netherlands with five euro-area countries that never changed their buffer (Austria, Finland, Italy, Malta, Luxembourg). The red line is the Dutch difference; the grey band shows differences that arise by chance when each control country is treated as if it had raised its buffer. Mortgage rates stayed inside the band, and household credit grew faster than in the controls, partly as a catch-up.

We find **no sign of contraction**, although this is not a causal estimate. DNB reached the same view in September 2026.

Source: ECB Data Portal (MIR, BSI); ESRB CCyB table (control selection); DNB press release 17 Sep 2026; own two-way fixed-effects event study with placebo inference.

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STORY · what to say

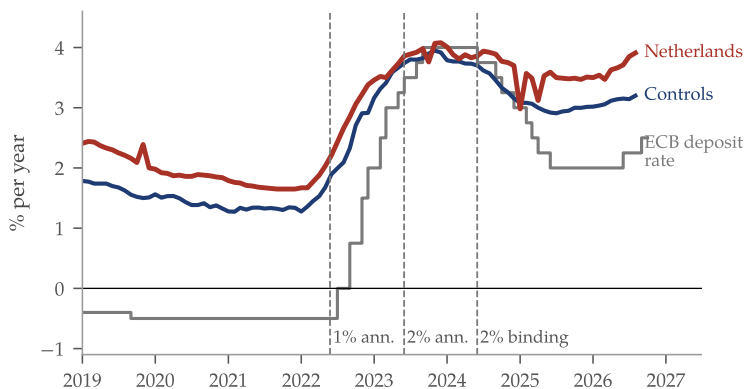
- Did borrowers pay instead? We compare Dutch lending with five euro-area countries that never changed their buffer: Austria, Finland, Italy, Malta and Luxembourg.
- The red line is the Netherlands minus those countries. The grey band shows the differences that arise by chance, when we pretend each control country was the one that raised its buffer. Mortgage rates stay **inside the band** at all three dates. Household credit actually grew **faster**: Dutch bank credit grew 16.6 percent from early 2022 to early 2026, against 7.9 percent in the euro area, partly as a catch-up after lagging before.
- So we find **no sign of contraction**. With only one treated country this is not a causal estimate, but DNB reached the same conclusion in September 2026.

LIKELY CHALLENGES · definitions and logic

- Q What is an event study?**
 - We track the difference between the Netherlands and control countries month by month before and after each CCyB date, with country and month fixed effects.
- Q What is a placebo test here?**
 - We rerun the estimate pretending each control country was treated. The range of these fake effects (the grey band) shows what chance alone produces.
- Q Why these five controls?**
 - Austria, Finland, Italy, Malta and Luxembourg did not change their CCyB in 2021–25, so they are clean. Countries that raised their own buffers would contaminate the comparison.
- Q Why is it not causal?**
 - One treated country, a small control group, a pre-announced path (anticipation) and the Dutch housing rebound. So we only say 'no sign of contraction'.
- Q Why did Dutch credit grow faster?**
 - Partly catch-up: in 2019–22 Dutch bank credit grew 4.3% vs 11.3% in the euro area; in 2022Q1–2026Q1 it grew 16.6% vs 7.9%.

Nor did the buffer double the ECB's squeeze. It let monetary policy focus on inflation

Q3 · IMPLICATIONS



A natural worry is that two tightenings at once, higher policy rates and higher capital requirements, would choke credit. The chart shows that Dutch mortgage rates followed the ECB's rate and moved closely with those in the control countries.

The ECB argues that early CCyB activation "helps monetary policy focus on its primary objective of price stability." (ECB, 2025)

With one monetary policy for twenty countries, the CCyB is the Dutch-specific lever, and it is insurance in case a rate shock goes wrong.

Source: ECB Data Portal (MIR mortgage rates; deposit facility rate); ECB Macroprudential Bulletin 31 (Detken, Hempell & Pirovano, 2025). Controls: AT, FI, IT, MT, LU.

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STORY · what to say

- A natural worry is that two tightenings at once, higher policy rates and higher capital requirements, would choke credit. The chart shows that this did not happen. Dutch mortgage rates, in red, followed the ECB's deposit rate, in grey, and moved closely with mortgage rates in the control countries, in blue.
- The ECB argues that building buffers early lets monetary policy focus on its main goal, price stability. With one monetary policy for twenty countries, the countercyclical buffer is the **Dutch-specific lever**. It is also insurance in case a rate shock goes wrong.

LIKELY CHALLENGES · definitions and logic

Q Macro- vs monetary policy?

- Monetary policy sets interest rates for price stability across the euro area. Macroprudential policy targets the stability of the financial system and can be set nationally.

Q Didn't both policies tighten at once?

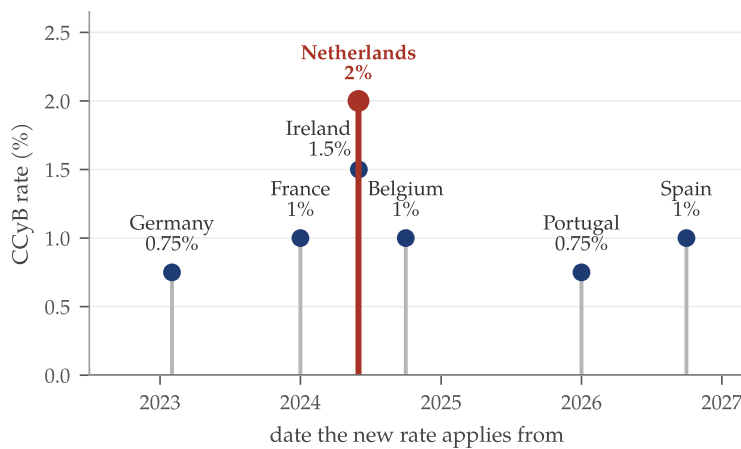
- Both tighten credit at the margin, but Dutch mortgage rates moved with the ECB rate and with the control countries, and bank profits rose with rates.

Q Why is a national lever useful in a currency union?

- The ECB cannot react to Dutch-specific financial conditions. The CCyB is the country-specific tool, and a buffer built early lets monetary policy focus on price stability (ECB, 2025).

Through reciprocity the 2% also binds foreign lenders, and the rest of Europe is moving the same way

Q3 · IMPLICATIONS



Under EU reciprocity, foreign banks that lend into the Netherlands must also hold the Dutch 2% on those loans. This limits leakage to foreign lenders, a known weakness of national capital rules, although lending can still move to non-banks.

The Netherlands was early and set the highest rate, but it was not alone. Since 2023 Germany, France, Ireland, Belgium, Portugal and Spain have all raised their buffers.

The Dutch move is part of a broader European shift towards keeping releasable capital in normal times.

Source: DNB press release 31 May 2023; ESRB CCyB rates table (Sep 2026); ECB Macroprudential Bulletin 21 (Behn et al., 2023); Aiyar, Calomiris & Wieladek (2014).
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STORY · what to say

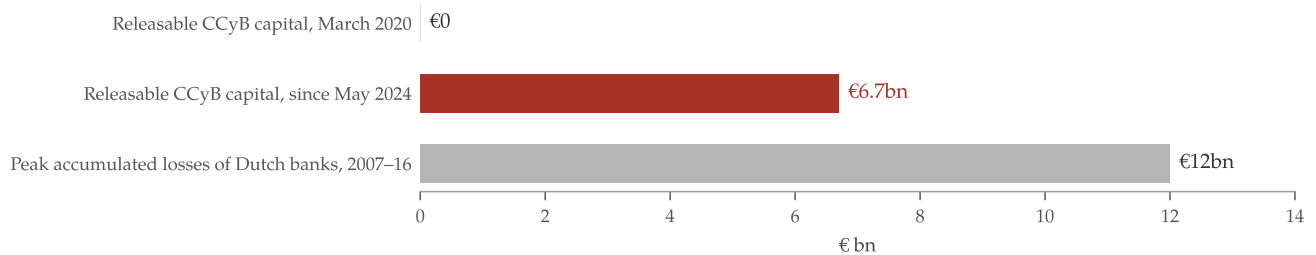
- The buffer also reaches across borders. Under EU reciprocity, foreign banks that lend into the Netherlands must also hold the Dutch two percent on those loans. This limits leakage to foreign lenders, which is a known weakness of national capital rules, although lending can still move to non-banks.
- And the Netherlands is part of a European shift. The chart shows the rates and the dates from which they apply. Germany, France, Ireland, Belgium, Portugal and Spain have all raised their buffers since 2023. The Netherlands was early and set the highest rate, but it was not alone.

LIKELY CHALLENGES · definitions and logic

- Q Define reciprocity.**
 - Foreign banks apply the Dutch CCyB rate to their Dutch exposures. In the EU this is mandatory up to 2.5%.
- Q Define leakage.**
 - Credit shifting to lenders not covered by the rule, such as foreign branches or non-banks. Reciprocity closes the foreign-bank channel; non-banks remain outside.
- Q Did lending move to non-banks?**
 - We have not measured non-bank mortgage shares, so we cannot claim either way. That is a limitation.
- Q Was the Netherlands an outlier?**
 - It was early and at the top (2%), but Germany, France, Ireland, Belgium, Portugal and Spain all raised their buffers since 2023.

The result: next time, DNB can release €6.7bn overnight

Q3 · IMPLICATIONS



In March 2020 the Netherlands had no releasable capital. Today it has €6.7bn, about half of the peak losses after the financial crisis and enough to support up to €150bn of lending. Research suggests the payoff is asymmetric: a one-point increase in normal times reduces lending by only about 0.1%, while a one-point release in bad times can raise it by up to 10%.

With the mortgage risk-weight floor expiring in November 2026, DNB says this “underpins the importance of . . . the current countercyclical capital buffer (CCyB) of 2%.”

The buffer is cheap to hold and powerful to release.

Source: DNB framework (2022); Lang & Menno (2023), ECB WP 2828; DNB press release 24 Apr 2026.
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STORY · what to say

- The result is shown in this chart. In March 2020 the Netherlands had no releasable capital at all. Today it has **6.7 billion euros**, about half of the peak losses after the financial crisis, and enough to support up to 150 billion euros of lending.
- This buffer is cheap to hold and powerful to release. Research on European banks suggests that a one-point increase in normal times reduces lending by only about 0.1 percent, while a one-point release in bad times can raise lending by up to ten percent.
- And its role is growing. The floor on mortgage risk weights expires in November 2026, and DNB says that this underlines the importance of the two-percent buffer.

LIKELY CHALLENGES · definitions and logic

Q Define a state-dependent effect.

- The impact of a policy depends on the state of the economy. Lang & Menno (2023): +1pp in normal times cuts lending by about 0.1%; a 1pp release in bad times can raise it by up to about 10%.

Q How much is releasable now?

- About €6.7bn (€3.3bn + €3.4bn), about half of the €12bn peak losses after the GFC, supporting up to €150bn of lending.

Q Why does the floor expiry matter?

- From 30 November 2026 the mortgage risk-weight floor no longer adds capital for housing risk; DNB says this ‘underpins the importance’ of the 2% CCyB.

The price is reliance on discretion, and the buffer has never been tested

Q3 · IMPLICATIONS

Credibility	Without a mechanical rule, markets must trust DNB's judgement, so DNB has to explain every decision. This is the classic trade-off between rules and discretion (Kydland & Prescott, 1977).
Calibration	Why 2% and not 1.5%? The ECB's loss-based estimates range from 1.1% to 1.8%, so the Dutch rate sits at the top of the banking union.
Untested	The buffer has never been released. Will banks lend the freed capital, or hoard it, as many did with usable buffers in COVID?
Baseline	Compared with 2021 rather than 2019, the refill <i>was</i> an increase: €3.3bn before the O-SII offset.
The same profits, twice	The ECB lists the Netherlands among the countries that adopted levies on banks' excess profits in 2022–23. These levies partly claim the earnings that fund the buffer.

None of these overturns the verdict, but they define what DNB must get right next time.

Source: Kydland & Prescott (1977); De Nora et al. (2025); ECB Macroeprudential Bulletin 21 (2023) and 31 (2025); DNB (2022, 2023).
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STORY · what to say

- Of course there is a price, and we see five critiques.
- **Credibility:** without a rule, markets must trust DNB's judgement, so DNB has to explain every decision. **Calibration:** why two percent, when the ECB's estimates range from 1.1 to 1.8? **Untested:** the buffer has never been released, and in 2020 many banks did not use the buffers they were allowed to use. **Baseline:** compared with 2021 rather than 2019, it was an increase. **Profits:** the same profits that fund the buffer were also targeted by levies on excess profits.
- None of these overturns our verdict, but they define what DNB must get right next time.

LIKELY CHALLENGES · definitions and logic

Q Rules vs discretion?

- Rules give predictability and credibility; discretion gives flexibility but risks inaction or time inconsistency (Kydland & Prescott, 1977). DNB's pre-announced path is a commitment device within discretion.

Q Will banks actually use released capital?

- Untested in the Netherlands. A released CCyB lowers the requirement itself, so it avoids the MDA stigma that stopped banks using buffers in 2020.

Q Isn't the baseline choice convenient?

- Yes, it matters: vs pre-COVID the refill is roughly neutral; vs 2021 it was an increase of €3.3bn before the O-SII offset. We show both.

Q What about excess-profit levies?

- The ECB lists the Netherlands among countries with levies on banks' excess profits in 2022–23, which partly claim the same earnings that fund the buffer.

In short, the promise was kept: the buffer was refilled in a storm, at the moment it was cheapest

CONCLUSION

Q1 · ENVIRONMENT

A recovered economy was hit by war, inflation and rate hikes. Credit was cold, housing was hot and then cooled, and the banks were strong and profitable.

Q2 · CRITERIA AND RISKS

A 2% normal level, calibrated on past losses, insures against shocks nobody can forecast. Housing went to other tools, and structural buffers were cut in return.

Q3 · IMPLICATIONS

The refill was cheap and largely a swap, with no sign of contraction. It complements monetary policy, binds foreign lenders and leaves €6.7bn releasable, at the price of discretion.

In March 2020, the Netherlands had nothing to release.
Next time, it will.

The best time to fill a buffer is when nobody thinks you need it, even in a storm.

STORY · what to say

- Let us conclude with the three questions of the assignment. The **environment** was a recovered economy hit by war, inflation and rate hikes, with cold credit, a cooling housing market and strong banks. The **criteria** were a two-percent normal level calibrated on past losses, as insurance against shocks nobody can forecast. The **implications**: the refill was cheap, largely a swap, with no sign of contraction, and it leaves 6.7 billion euros ready to release, at the price of more discretion.
- *(pause)* In March 2020 the Netherlands had nothing to release. **Next time, it will.** The best time to fill a buffer is when nobody thinks you need it, even in a storm. Thank you, and we look forward to your questions.

LIKELY CHALLENGES · definitions and logic

- Q What is your main contribution?**
 - Showing, with a testable comparison, that the Dutch increase was a refill of releasable capital rather than a response to a credit boom, and that it was cheap when it happened.
- Q What would change your verdict?**
 - DNB going above 2% while credit indicators rise, or evidence that lending contracted around the binding dates.
- Q Would you recommend this to other countries?**
 - Where the gap is unreliable and buffers are mostly structural, yes: build a releasable buffer in normal times, gradually, and offset structural charges. The cost is greater reliance on judgement.

Glossary

APPENDIX

CCyB	Countercyclical capital buffer: 0–2.5% of risk-weighted assets on domestic exposures; raised with a 12-month lag, released immediately.	Credit-to-GDP gap	Credit/GDP ratio minus its long-run trend; the Basel reference for the CCyB (gap 2–10pp → buffer 0–2.5%).
Positive neutral CCyB	A CCyB kept above zero when risks are neither high nor low, so it can be released for any shock.	HP filter	Hodrick-Prescott filter: estimates a smooth trend; one-sided uses only past data (real time), two-sided also future data.
CET1	Common Equity Tier 1: the highest-quality capital (shares, retained earnings), as % of RWA.	Headroom	CET1 capital above a bank's requirement / MDA trigger.
RWA	Risk-weighted assets: assets weighted by their riskiness (e.g. mortgages count less than corporate loans).	Reciprocity	Foreign banks must apply the Dutch CCyB rate to their Dutch exposures.
Pillar 1 / P2R / P2G	Legal minimum (4.5% CET1) / bank-specific add-on set by the supervisor / non-binding supervisory guidance.	Borrower-based measures	Limits on loan-to-value (LTV) and loan-to-income (LTI) of new mortgages.
CCoB	Capital conservation buffer, 2.5% for all banks.	Risk-weight floor	Minimum average risk weight on Dutch mortgages of banks using internal models (Art. 458 CRR), Jan 2022 – Nov 2026.
O-SII / SRB	Buffer for other systemically important institutions / systemic risk buffer (structural, not designed for release).	Event study / placebo	Compare NL with countries that did not change their CCyB around each date / repeat with each control as a fake “treated” country.
MDA	Maximum distributable amount: if CET1 falls below the combined buffer requirement, payouts are automatically restricted.	pp / bp	Percentage point / basis point (0.01pp).

STORY · what to say

- Backup only. Use it when a question involves a term; point to the row and give the one-line definition.

LIKELY CHALLENGES · definitions and logic

- Q Use this page to define any term quickly.**
- Point to the term and give the one-line definition; then return to the slide being discussed.

Methods and references

APPENDIX

Methods. Credit gap: one-sided (recursive) and two-sided HP filter, $\lambda = 400,000$, sample from 1961Q1; Hamilton (2018) regression filter, $h = 8, 20, p = 4$. Headroom: (CET1 ratio – CET1 requirement/MDA) $\times$ total RWA per bank; CCyB amount sector-wide (upper bound); 6–47% assumption grid as sensitivity. Event study: two-way FE, NL $\times$ 3-month event-time bins, country and month FE; placebo inference over controls with no CCyB change 2021–25. Code and data: GitHub repository (code/, data/).

Primary sources

BCBS (2024). Range of practices in implementing a positive neutral CCyB (d585).
 DNB (2020). DNB lowers bank buffer requirements to support lending, 17 Mar.
 DNB (2022). Analytical framework for setting the CCyB in the Netherlands.
 DNB (2022, 2023). CCyB press releases, 27 May 2022, 31 May 2023; O-SII news item, 31 May 2023.
 DNB (2026). CCyB decisions 26 Mar and 17 Sep; risk-weight measure expires, 24 Apr.
 DNB. Financial Stability Report, spring 2022 and spring 2023.
 Knot, K. (2023). Introductory statement, Standing Parliamentary Committee for Finance, 7 Jun.
 ECB (2020). Temporary capital and operational relief, 12 Mar; ECB (2022). Monetary policy decisions, 21 Jul.
 ESRB. CCyB rates table; O-SII notifications NL (27 Nov 2020; 31 May 2023); COVID-19 measures.
 Data: BIS (credit gaps, total credit, DSR, property prices); ECB Data Portal (MIR, BSI); CBS; bank annual and Pillar 3 reports (ING, Rabobank, ABN AMRO, de Volksbank).

Literature

Aiyar, Calomiris & Wieladek (2014). Does macro-prudential regulation leak? *JMCB* 46(s1).
 Behn, Pereira, Pirovano & Testa (2023). A positive neutral rate for the CCyB. *ECB MPB* 21.
 Couaillier, Lo Duca, Reghezza & Rodriguez d'Acari (2022). Caution: do not cross! *ECB WP* 2644.
 De Nora, Pereira, Pirovano & Stammwitz (2025). From losses to buffer. *ECB WP* 3061.
 Detken, Hempell & Pirovano (2025). Macroprudential and monetary policy interaction. *ECB MPB* 31.
 Drehmann, Borio, Gambacorta, Jiménez & Trucharte (2010). Countercyclical capital buffers: exploring options. *BIS WP* 317.
 Edge & Meisenzahl (2011). The unreliability of credit-to-GDP ratio gaps in real time. *IJCB* 7(4).
 Hamilton (2018). Why you should never use the Hodrick-Prescott filter. *REStat* 100(5).
 Herrera, Scalone & Pirovano (2024). The importance of being positive. *ECB MPB* 24.
 Kydland & Prescott (1977). Rules rather than discretion. *JPE* 85(3).
 Lang & Menno (2023). The state-dependent impact of changes in bank capital requirements. *ECB WP* 2828.
 Repullo & Saurina (2011). The countercyclical capital buffer of Basel III: a critical assessment. *CEMFI WP* 1102.
 Tinbergen (1952). *On the Theory of Economic Policy*. North-Holland.

STORY · what to say

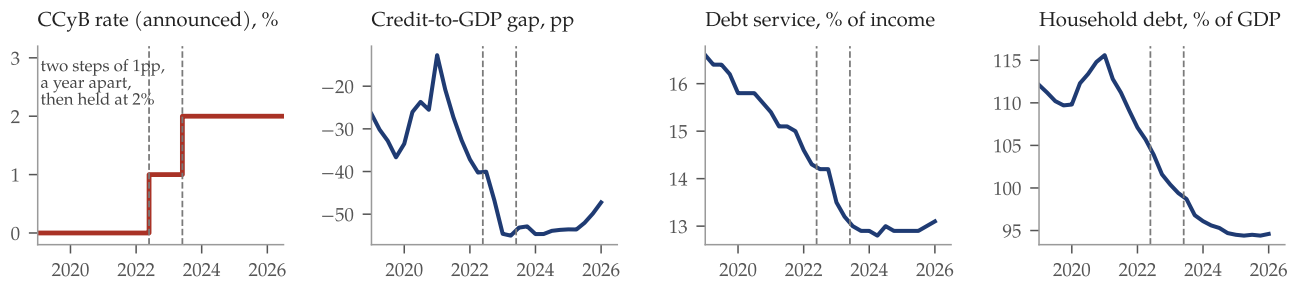
- Backup for method questions.
 - **Credit gap:** one-sided and two-sided HP filter ($\lambda = 400,000$), plus Hamilton regression filter.
 - **Headroom:** (CET1 ratio minus requirement) $\times$ total RWA, per bank.
 - **Event study:** two-way fixed effects, three-month bins, placebo inference over five never-treated countries.
- All references are verified.

LIKELY CHALLENGES · definitions and logic

- Q **How exactly is the event study specified?**
 - Two-way fixed effects: outcome on NL $\times$ 3-month event-time bins, with country and month fixed effects; placebo inference over clean controls (AT, FI, IT, MT, LU).
- Q **How is headroom computed?**
 - (CET1 ratio – CET1 requirement/MDA trigger) $\times$ total RWA per bank; the CCyB amount is sector-wide, so shares are upper bounds.

Appendix A1 · The verdict, rows 1 and 2: the buffer rose in two fixed steps while every credit indicator fell

APPENDIX



Row 2 · THE PATH

Normalisation predicts one percentage point a year and a stop at 2%. DNB announced 1% in May 2022 and 2% in May 2023, and has held 2% ever since.

The buffer moved on a timetable, not with credit.

Row 1 · CREDIT INDICATORS

A boom fighter raises the buffer when credit indicators rise. Between the two announcements (dashed lines) the gap, the debt-service ratio and household debt all fell.

Source: ESRB CCyB rates table; DNB press releases (27 May 2022, 31 May 2023); BIS (WS_CREDIT_GAP, WS_DSR, WS_TC). Dashed lines: CCyB announcements.
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STORY · what to say

- The left panel shows the path: two steps of one point, a year apart, then held at two percent, exactly as normalisation predicts.
- The other three panels show the credit indicators. Between the two announcements the credit gap, the debt-service ratio and household debt all fell, so the buffer did not follow credit.

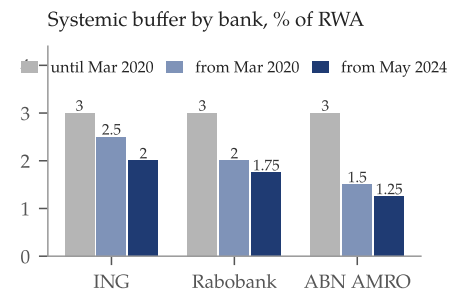
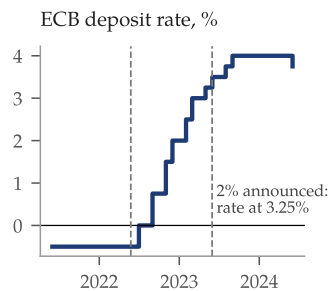
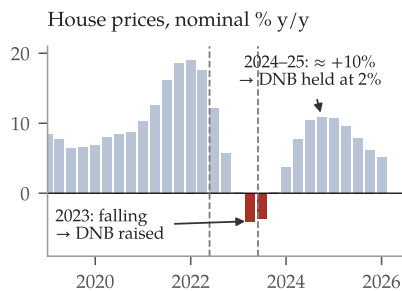
LIKELY CHALLENGES · definitions and logic

Q Why does the path matter?

- A boom fighter's buffer follows risk indicators; a normalising buffer follows a pre-announced timetable. The CCyB moved in two equal steps a year apart while the gap, debt service and household debt all fell.

Appendix A2 · The verdict, rows 3 to 5: DNB raised into falling prices, held when prices boomed, and cut other buffers

APPENDIX



Row 3 · 2023

Prices were falling and the ECB rate stood at 3.25%. A boom fighter would pause; DNB raised.

Row 4 · 2024–26

House price growth returned to about 10% a year. A boom fighter would go above 2%; DNB held at 2%.

Row 5 · OTHER BUFFERS

DNB cut the large banks' systemic buffers in 2020 and again in 2024.

All five rows point the same way: **normalisation, not tightening.**

Source: BIS (WS_SPP); ECB Data Portal (deposit facility rate); ESRB O-SII notifications (2020, 2023); DNB (17 Mar 2020).
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STORY · what to say

- On the left, house prices: in 2023 they were falling, yet DNB announced two percent. In 2024 and 2025 they grew about ten percent a year, yet DNB held at two percent.
- In the middle, the ECB deposit rate stood at 3.25 percent when the second step was announced.
- On the right, the systemic buffers of the three large banks were cut in 2020 and again in 2024, which is what normalisation predicts.

LIKELY CHALLENGES · definitions and logic

- Q Why show house prices and the ECB rate?
 - They are the two conditions under which a boom fighter would have behaved differently: pause in 2023 when prices fell and rates jumped, go above 2% in 2024–25 when prices rose about 10%.
- Q What does the right panel prove?
 - That structural buffers were cut twice (2020 and 2024), which only makes sense if the CCyB was replacing them rather than adding to them.